Towards Personalised Support for Monitoring and Improving Health in Risky Environments

Jayalakshmi Baskar and Helena Lindgren
{jaya,helena}@cs.umu.se

Department of Computing Science, SE-901 87 Umeå, Sweden

Abstract. The research presented in this paper aims at supporting an individual in their daily living in a home environment. Our objective is to fulfill the scenario of an individual older adult having breakfast and who would like to have social interaction and be aware of happenings outside of her home. This is accomplished by extending the existing ACKTUS ontology with an ubiquitous perspective for activity recognition and secondly, to tailor support and provide personalised services to an older adult in home environment based on activity recognition. This is done by means of an agent system, enabling dialogues between agents and between the human and a software agent.

1 Introduction

There is a need for methods for software agents to reason about risks and risky situations, based on synthesised information obtained from different sources of information in a ubiquitous environment [1, 2]. This is for the purpose to make decisions about what advices to provide the user, or in which way to adapt the environment to the user's needs. These decisions should be founded in the obtained information specific to the user combined with generic knowledge about hazardous factors in an environment. Moreover, in order to use the intelligent environment for empowering the user, there is also a need for methods to allow the user to maintain the power over the intelligent environment [3].

The presented research addresses adaptive computer-based personalised support for monitoring and improving health of older adults for the purpose to lead an independent life. In order to achieve adaptability of personalised information, it is important to observe the user's behaviour, and make predictions based on those observations. The information pertaining to individual user obtained from such observations is known as an *user model* (e.g., [4]). Personalisation aims at providing users with the content that they need without necessarily requiring the users to specify it explicitly [5]. Thus categories of knowledge needed to provide personalised support for activities are knowledge about *the user*, *the user's activities and environment*, and *generic domain knowledge*. The knowledge about the user consists of the individual's preferences [6], ability, interests, habits, needs, wishes and social network, etc. The knowledge about the user's activities deals with what is being done, its purpose and how a particular activity is performed. The third category is environmental knowledge that concerns physical, social and virtual components. To a certain extent these types of knowledge can be obtained by a system through *information-seeking dialogues* between the user and the system agent [7]. This method can be supplemented with observations of the user in activity. The user may describe explicitly what activity is being performed and their environment, while the latter involves observation of both environment and activity, e.g., by the use of sensors as part of a ubiquitous computing environment [8, 10, 11]. The user, activity and environment knowledge is specific to an individual whereas the last category, the domain knowledge is generic and created by domain experts such as a health care professional based on reliable knowledge-sources [12].

The main contribution of this work is a developed terminology model used for agents' reasoning about activities performed in an ambient assisted living (AAL) home environment. Moreover, an initial implementation of an activity recognition system and a multi-agent dialogue system reasoning about activities are presented. A pilot study with the purpose to investigate needs and motives of older adults related to accomplishing activities they wanted to perform, generated a generic model of purposeful activity [13]. The following activities were identified: 1) feeling safe and secure, 2) having knowledge and control (e.g., keeping up with news), and 3) feeling good, engaged, active, having fun. The second result i.e., keeping up with news and third to be socially active have been a motivation for this research work.

The results consist of the following: 1) an extension of the ACKTUS core ontology for the purpose to capture activities in a physical and virtual environment, 2) initial design and implementation of the MUDRA system for activity recognition and evaluation, and 3) an initial design and implementation of an agent dialogue system where agents communicate with the end user in a home environment.

The paper is organized as follows. In Section 2 the ACKTUS tool is introduced and an architecture of the system for personalised support in home environment with a case scenario is described. In Section 3 the extension of ACKTUS ontology from ubiquitous computing perspective is presented. In Section 4, activity recognition has been discussed with a brief explanation about MUDRA system for activity recognition, followed by multi-agent dialogues and some conclusions and directions for future work.

2 System architecture

The ACKTUS (Activity Centered Knowledge and interaction modeling Tailored to Users) platform is used by domain professionals for modeling the domain knowledge, user models and personalised behavior of a knowledge-based support system [14]. Such system may utilize sensor information from an ambient assisted living (AAL) environment for the categorisation and interpretation of observations in order to represent knowledge about recognisable purposeful activities, useful for support provision. The system architecture is shown in Figure 1. ACKTUS-care and a part of the functionality of I-Help are developed using the ACKTUS platform, which domain expert physicians can use for modifying the knowledge content and interaction design [14, 15].

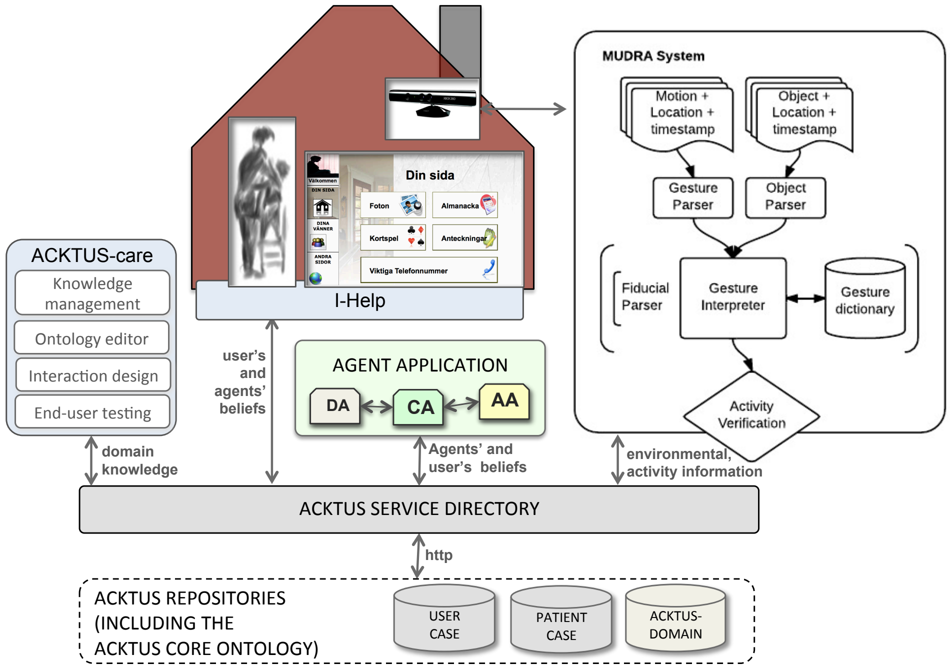

Fig. 1. Architecture.

2.1 Rut Scenario

Our scenario is based on an older woman who suffered from a few falls before a hip fracture [12]. Part from having pain when walking around, she is also very worried that she might fall again. She has mild cognitive dysfunction, showing that she sometimes forgets to take medication and eat properly. She likes to lead an independent life without a caretaker and thus she needs a solution to monitor and improve her health along with fulfillment of her wishes and needs of daily living.

3 Ontology

For representing the necessary knowledge to be used in reasoning about tailored support in an intelligent environment, we extend the ACKTUS ontology, which contains a basic model of the user and activity [15]. ACKTUS is based on models of human occupation [16], the International Classification of Disability, Functioning and Health (ICF) provided by the World Health Organization [17], and other medical terminologies. We take a persona and a case scenario as starting point for the development of an individual's supportive knowledge-based environment [12]. It is then used to extend the *ACKTUS user model*, to create an *environment ontology* and *activity ontology*. Activity theory has been used to categorize and interpret activities in order to represent knowledge about recognizable activities [16].

In order to create a holistic understanding of Rut's environment as an adaptable and adaptive AAL environment, we evaluated the conceptual models underlying the two frameworks ACKTUS and egocentric interaction [8]. The selected part is the breakfast moment, specified by Rut's physical presence during a time frame from when she enters the dining area to when she leaves the area. Based on this the ACKTUS ontology was extended to include the class *space*, which can be both physical and virtual. Spaces are locations where objects and actors are situated during activities. Features that relate to the physical home environment (located in spaces) are captured by the ACKTUS ontology at a generic level using the *building* class (Figure 2). In our example we identify the particular sub-space in the kitchen as dining area.

Activities fall under the *Activities and Participation* class, following the ICF categorisation as basic structure (Figure 2). The activities that are performed in the space in focus during Rut's breakfast moment are done partly in parallel and are the following: having breakfast (includes actions such as eating, drinking, pouring water into the coffee cup, and milk into the porridge plate and a glass); keeping up with society by reading news, keeping up with her social network by having a dialogue with her friend Greta; caring for her health by taking her medication; train and get some entertainment by playing with cognitive training tool; plan the day by checking her agenda/schedule; and finally, in order to reflect upon daily life she has a weekly dialogue with the *Coach Agent* (CA) (refer Section 4.2) of the AAL environment. In the initial assessment phase of our scenario, the motives behind these activities were identified by the occupational therapist, Rut and her son, and the activities were selected as being important activities that Rut wanted to have support with. The dialogue with the CA serves as a followup of how Rut thinks that her daily life works, and of her emotional state (e.g., Figure 4). This subjective view is to become supplemented in this project with an activity recognition functionality.

Some of these activities are taking place in both a physical space (dining area) and in virtual spaces (a news forum on the web, a social media application that gives access to her friend and her home, a web-based cognitive training tool, etc) making these activities taking place in a *mixed-reality space*. This characteristic feature is also captured by the egocentric conceptual model. A significant difference between the models is that while the ACKTUS ontology covers purposeful activities, their tools, motives, actors and locations, as how Rut and health professionals perceive activity, the egocentric conceptual model covers primarily concepts related to sensors, effectuators, interaction devices and

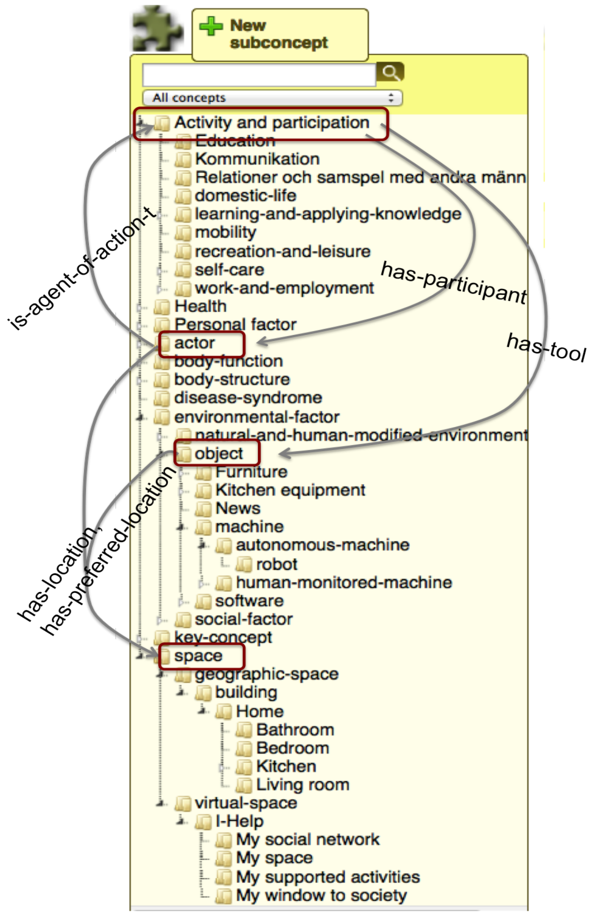

Fig. 2. An overview of the nodes in the extended core ontology that builds the user, activity and tool models. Some of the added classes and their properties are shown, e.g., space and its subclasses, and additional objects relevant in a home environment.

their properties and their relation to the situative spaces [8]. This means that focus lies on the operational level of activity, e.g., how accessible objects are to the actor and through which modalities information is communicated. In this way the two conceptual models are supplementary. e.g., Rut's preferences regarding how to interact with the systems captured at the purposeful level of activity and by the ACKTUS model can be effectuated by the egocentric interaction framework.

4 Activity recognition

A pilot study was conducted to investigate in which way gesture based interaction could be a means for performing activities. Another purpose was to develop a system, which can form the base for activity recognition and evaluation both in home and work environments. A hierarchical task analysis was done for the purpose to identify suitable tasks to be identified using 3D sensors (refer Figure 3). Hierarchical task analysis (HTA) is a method that was developed to understand how a specific process is performed [9], in a step by step manner. The task analysis was based on the use case scenario and the persona's needs and wishes. The tasks, which were assumed suitable to be identified through sensors, were studied in an empirical study involving a convenient sample of 20 individuals (10 male and 10 female) with the purpose to explore which gestures are most natural for each task. The results of this pilot study form the basis for the development of a prototype system MUDRA (Machine Understandable Decision about Recognisable Activity) for activity recognition based on gesture and object identification.

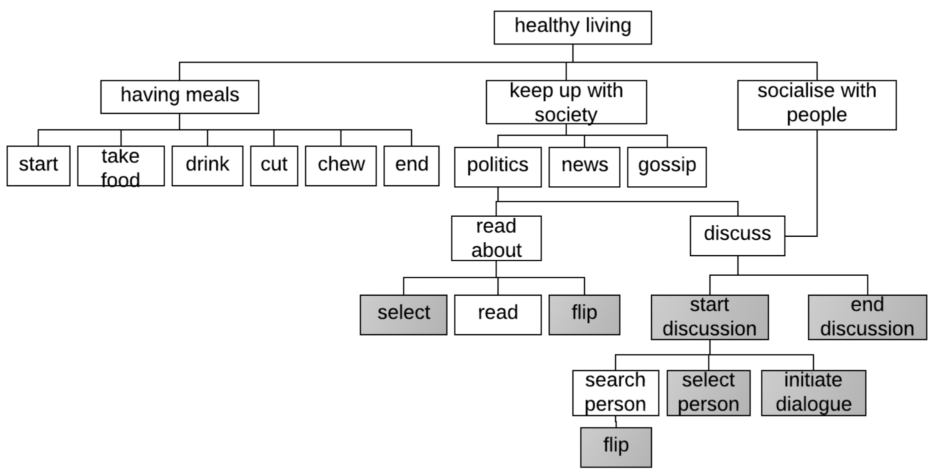

Fig. 3. Part of the task analysis is shown with the selected tasks to be identified by a gesture recognition system highlighted in darker color.

4.1 The MUDRA system for activity recognition

We take as starting point the selected activities, which the user prioritizes as important and which the user wants to have support with. Consequently, we also need to identify activities. For this purpose, an activity recognition system is being developed, based on 3D sensors. Since the core task is to detect movement of limbs, we start by performing a pilot study to find out what types of gesture based interaction could be used for performing activities. By developing gesture recognition and evaluation, a base for detecting also other movement characteristic for activities is formed.

The prioritized activities in focus for activity recognition as part of this work, are keeping up with what is happening in society by reading the news, and at the same time, having a conversation with a friend about the news through a social network application. The targeted actions to be detected in the initial phase of our work, were *select* person to communicate with, *start* and *end* a dialogue. Moreover, the task to *flip* (e.g., for changing page of a news paper, or scroll between pages in an application) was included.

In the breakfast scenario, there are two types of interaction identified 1) interaction with a digital portrait, which mediates a social network functionality integrated in I-Help (Figure 1), and 2) interaction with kitchen objects such as a cup for accomplishing the task of having breakfast. Focus has been on development of algorithms to *detect and identify gestures and objects* and their manipulations. Moreover, *activity verification and evaluation* involves distinguishing between similar gestures used for different purposes, e.g., social activity conducted in parallel to having breakfast and reading news. MUDRA allows an older adult to interact with a digital portrait on the wall, with natural hand gestures. The purpose is to select a person in her social network and initiate a conversation with this person. When MUDRA has recognized the gesture as a dialogue control maneuver, the corresponding command is sent to the social network application. In other cases, the interpretation of activity is negotiated as part of an agent dialogue procedure involving the Activity Agent (AA) and the Coach Agent (CA).

4.2 Agent-based dialogues about activities in a home environment

For accomplishing reasoning and dialogue-based support we develop a multi-agent dialogue system, which enables dialogues between a human and software agents. It essentially follows the outline in [12] including the following software agents: a Domain Agent (DA), which uses a domain knowledge repository as belief base, and a Coach Agent (CA), which manages the user model by reasoning with other agents about conflicting interpretations of the human activities, and preferences obtained from different sources [18]. It is also the CA's task to protect the user's interests in dialogues with other agents. For the purpose of this work, human-agent dialogues are viewed as a means to providing the user control over the AAL environment, which is a goal for the work. Therefore, we design and implement the human-agent dialogue system partly for this purpose.

The goal of the multi-agent system is to provide analyses at two levels. Firstly, to detect and identify activity based on sensor data about objects and body movements, and on a user profile containing activity patterns and habits. Second, the identification of activity needs to be related to level of importance to the user, user's preferences and goals, in order to reason about how the system should act upon the new knowledge, e.g., providing advice, support in the form of new tasks to be performed, etc. In this section we provide initial results feeding into a system which can accomplish tailoring of the support provided an individual.

An *Activity Agent* (AA) is introduced, which collaborates with the CA and DA agents (Figure 1). The purpose of the AA is to collect information about observed activities obtained from different sources. In our system it is the MUDRA system which provides the environmental information and information about performed activities.

5 Agent dialogues for reasoning about purposeful activities

The user may control the AAL environment by explicitly expressing e.g., preferences, priorities, interests, and adjust these over time. The base for interaction is implemented as information-seeking dialogues as defined in [7], through which the user can inform the system about preferences, priorities, interests, etc. Based on the information the system provides feedback to the user in the form of: 1) suggestions of decisions, 2) advices, and 3) suggestions of actions to make for obtaining more knowledge about a situation. Currently, the actions are formally defined as ACKTUS assessment protocols. The information capture, inferences and responses are built upon knowledge and interaction flows modeled by the domain professionals.

Domain experts modeled the domain knowledge, user models and content of the dialogues required to realize our use case scenario (e.g., Figure 4). The domain experts could also model the flow of dialogues. For illustrating the dialogues in the development sessions with domain experts, we used an algorithm for executing, or simulating, the different types of dialogues. The algorithm makes use of ACKTUS assessment protocols, their content, their associated rules and their consequents as dialogue flows the way they were structured by the domain experts in the modeling sessions. In this highly structured form, the autonomy of agents is limited to a minimum. However, it serves as a starting point for our user-driven approach to development of agent-based dialogues. The domain professionals were, at the point of modeling the dialogues, in control of the agent's behavior. The same building blocks created using ACKTUS are used in the agent-based dialogues implemented as part of this work.

Important activities that the older adult wants to be able to perform in a satisfactory way can be identified and selected in the initial assessment and added into their user profiles. This is accomplished through the information-seeking dialogue systems described earlier. This selection will be used in a training phase of the activity recognition system. The information-seeking dialogue system implementing ACKTUS dialogues is being extended with a dialogue system where the user can interact with the system agents, proposing supportive and contradictory arguments. The initial design of this extension has been created, and the implementation is ongoing. The dialogues between human and software agents aim at providing a natural dialogue with the system. Therefore, the interaction design of the dialogues will be an important issue to explore in future work.

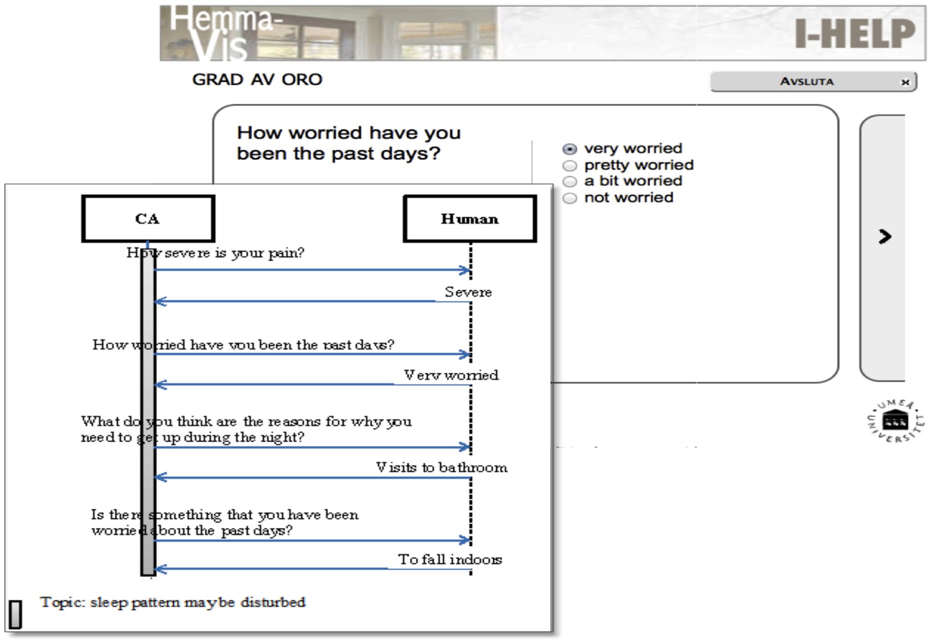

Fig. 4. Dialogue example of a weekly follow-up dialogue between CA and Rut.

5.1 Dialogue examples

Consider the case when Rut has had her breakfast one morning, and AA detects the following:

– Breakfast is completed
– Rut has moved away from her kitchen table without her walking aid
– Rut has not interacted with the pill dispenser
– Rut has had a conversation with her friend
– Rut has read the news

AA concludes the following:

– Rut has not taken her medication
– Rut is walking around in an unstable manner

The information is entered into Rut's profile, which causes a conflict with other information, which CA detects. CA initiates a dialogue with AA with the topic *"Rut has taken her pills"*, since this is the information CA has as default assumption based on Rut's testimony. AA attacks this with the argument: *"Rut has not taken her medication, based on the observation that she has not interacted with the pill dispenser"*.

This causes CA to initiate a dialogue with Rut with the same topic. CA poses the question about whether Rut has remembered to take her pills today? Rut responds that she thinks so, but checks the pill dispenser, and discovers that the pills for this morning is still there. Then she confirms that she had not, but that she will take them at that point instead.

Another topic, this time initiated by AA, is concerning the risk for falling. Since Rut is worried about falling, this is a prioritized topic in her case. Therefore, AA initiates a dialogue with CA and claims that Rut is walking around in an unstable manner. Since CA has the belief that Rut is sufficiently stable only with her walking aid, and that Rut has testified that she always uses the walking aid, CA argues against AA's claim. However, since AA builds its claim on the observation that Rut and the walking aid is not at the same place, CA again turns to Rut to resolve the conflicting views. The reasons for why Rut is not using the walking aid can be that she forgets to use it, or that there is something wrong with it. Another reason may be that she feels better, and more stable than before. To determine whether her experience matches her actual state a renewed assessment by a therapist is required, who considers a potential decrease in judgment ability, possible due to medication or an evolving dementia disease.

6 Conclusion and future work

The knowledge needed for the provision of personalised support for activities in an AAL environment consists of the individuals preferences, ability, interests, habits, needs, wishes, activities and social network. The goal of this work was to develop a model used for agents' reasoning about activities performed in an ambient assisted living (AAL) home environment.The results include an ACKTUS ontology extended with environmental and activity information, an initial design of a multi-agent dialogue system and of the activity recognition system - MUDRA, based on 3D sensors. Future work involves an extension of MUDRA from home environment to work environment where a support application for risk assessment of hazardous work environments will be integrated with sensor data in the mining work environment. By feeding more person-specific information into the algorithms we expect that the accuracy of the activity recognitions and risk assessments will increase. As an integral part of the development, end users will be involved in the design process and in evaluation studies.

References

1. Isern, D., Sánchez, D., and Moreno, A. Agents applied in health care: A review. I. J. Medical Informatics 79(3): 145-166 (2010)
2. Pollack, M.E., Brown, L., Colbry, D., Orosz, C., Peintner, B., Ramakrishnan, S., Engberg, S., Matthews, J., Jacqueline, D.J., McCarthy, C., Thrun, S., Montemerlo, M., Pineau, J., Roy, N.: Pearl: A Mobile Robotic Assistant for the Elderly, AAAI Workshop on Automation as Caregiver (2002)
3. Rogers, Y.: Moving on from weisers vision of calm computing: Engaging ubicomp experiences. In: P.Dourish, A.Friday (eds.) Ubicomp, Lecture Notes in Computer Science, vol. 4206, pp.404-421. Springer (2006)

4. Kobsa, A. Generic user modeling systems. User modeling and user adapted interaction, 11, 4963, (2001)
5. Alatalo, T., and Siponen, M. T. Addressing the personalization paradox in the development of electronic commerce systems. In Post-proceedings of the EBusiness Research Forum (eBRF), Tampere, Finland, (2001)
6. Oppermann, R. (ed.): Adaptive User Support Ergonomic Design of Manually and Automatically Adaptable Software. Computers, Cognition, and Work, Vol. 40, Lawrence Erlbaum Asociates, Hillsdale, NJ (1994)
7. Walton, D.N., Krabbe, E.C.W. Commitment in dialogue: Basic concepts of interpersonal reasoning. SUNY Press (1995)
8. Surie, D., Pederson, T., Lagriffoul, F., Janlert, L.E., Sjölie, D. Activity Recognition using an Egocentric Perspective of Everyday Objects. In: proc. of the IFIP International Conference on Ubiquitous Intelligence and Computing LNCS 4611:246-257 Springer (2007)
9. Stanton, N.A. Hierarchical task analysis: Developments, applications, and extensions. Applied Ergonomics. 37(1), 55-79, (2005)
10. Surie, D., Lagriffoul, F., Pederson, T., Sjölie, D.: Activity recognition based on intra and extra manipulation of everyday objects. Proceedings of the 4th international conference on Ubiquitous computing systems. Tokyo, Japan: Springer-Verlag (2007)
11. Patterson, D.J., Fox, D., Kautz, H., Philipose, M.: Fine-Grained Activity Recognition by Aggregating Abstract Object Usage. Proceedings of the Ninth IEEE International Symposium on Wearable Computers. IEEE Computer Society (2005)
12. Lindgren. H., Surie, D., Nilsson, I.: Agent-supported assessment for adaptive and personalized ambient assisted living. In: J. Corchado, J. Prez, K. Hallenborg, P. Golinska, R. Corchuelo (eds.) Trends in Practical Applications of Agents and Multiagent Systems, Advances in Intelligent and Soft Computing, vol. 90, pp. 25-32. Springer Berlin Heidelberg (2011)
13. Lindgren, H. Personalisation of Internet-Mediated Activity Support Systems in the Rehabilitation of Older Adults A Pilot Study. In: proc Personalisation for e-Health 2009, pp. 20-27, Verona, (2009)
14. Lindgren H, Winnberg PJ, Winnberg P.: Domain Experts Tailoring Interaction to Users - an Evaluation Study. In P. Campos et al. (Eds.): INTERACT 2011, Part III, LNCS 6948, pp. 644-661, Springer (2011)
15. Lindgren H, Winnberg P (2010) A Model for Interaction Design of Personalised Knowledge Systems in the Health Domain. In M. Szomszor and P. Kostkova (Eds.): E-Health 2010, LNICST 69, pp. 235-242, Springer Verlag (2011)
16. Kaptelinin, V.: Computer-mediated activity: Functional organs in social and developmental contexts. In: B. Nardi (ed.) Context and Consciousness. Activity Theory and Human Computer Interaction, pp. 45-68. MIT Press (1996)
17. International Classification of Functioning, Disability and Health (ICF), World Health Organization (WHO), [online] , http://www.who.int/classifications/icf/en/index.html (Accessed: 7 March 2013)
18. Lindgren, H., Yekeh, F., Baskar, J., Yan, C.: Agent-supported assessment for personalised ambient assisted living. In: Proc. Agents Applied to Health Care (A2HC 2012) (2012)